

HAT-P-27b

R-2178

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-27_260707 · created 2026-08-13T01:11:43+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2461228.7029 +/- 0.0091 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.111 +/- 0.04
Transit depth (Rp/Rs)^2	1.24 +/- 0.89 [%]
Orbital Inclination (inc)	86.62 +/- 2.89
Airmass coefficient 1 (a1)	3.00833 +/- 6e-05
Airmass coefficient 2 (a2)	-0.8 +/- 0.48
Scatter in the residuals of the lightcurve fit is	0.0018 %
Best Comparison Star	#1 - [627, 127]
Optimal Aperture	2.24
Optimal Annulus	5.74
Transit Duration (day)	0.0011 +/- 0.001

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.111 $\pm$ 0.04 vs 0.119 (deviation 0.2 σ)
Duration fit vs published	0.0011 d vs 0.07 d (deviation 68.9 σ)
Mid-time O-C	2.9 min
Depth SNR	2.8
Residual scatter	0.0018 %

Light curve

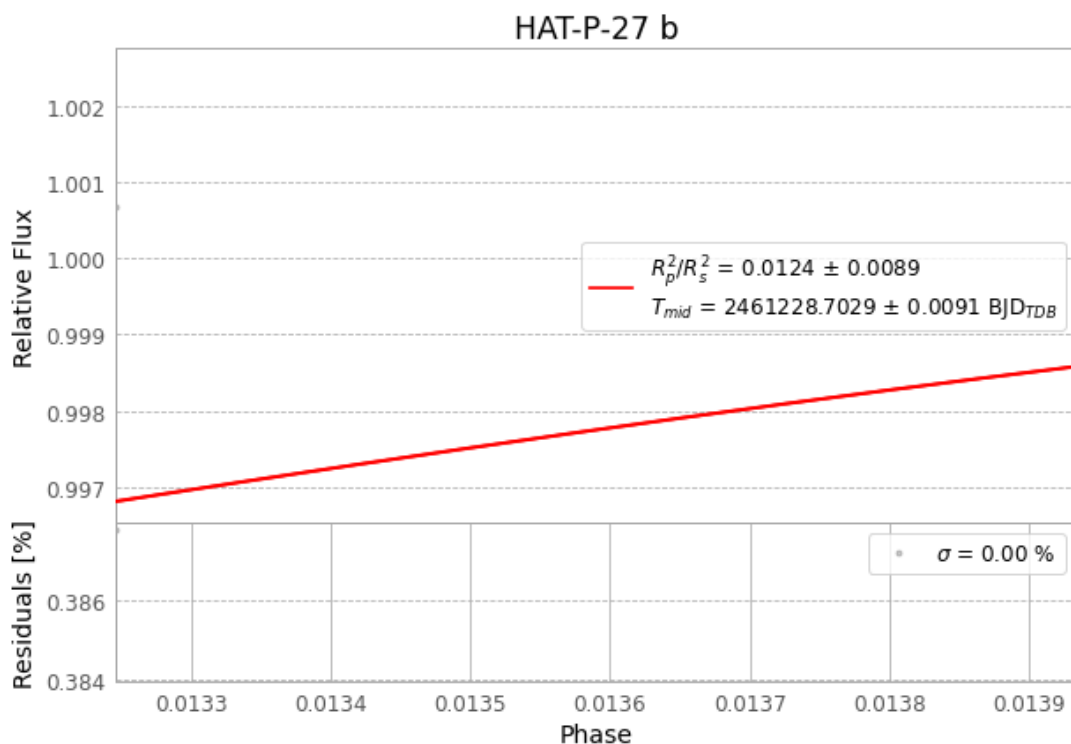


Figure 1 — FinalLightCurve_HAT-P-27 b_2026-07-06.png (SHA-256 9adf07a27117c414...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [560, 146], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 bc28102145e9050d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

125 FITS frames (48 MB), HATP-27260707025559.FITS.gz → HATP-27260707063616.FITS.gz. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-27-260707.log (a8c48d807402...), FinalLightCurve_HAT-P-27 b_2026-07-06.png (9adf07a27117...), FinalLightCurve_HAT-P-27 b_2026-07-06.csv (226299c60189...)

Compute resources

Wall time 107.6 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-13 01:11Z.